

DUNCAN TAI BENNETT

Data Scientist | ML Engineer | PhD Mathematician

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🏢 PROFESSIONAL EXPERIENCE

University of Arizona

Aug. 2019 – Aug. 2025

Graduate Research Assistant

Tucson, AZ

- Conducted research at the intersection of machine learning, dynamical systems and scientific computing to develop computational methods for the computer-assisted proof of chaos in the Long Short-Term Memory model.
- Developed numerical simulations and rigorous computations in Python, Matlab and C using interval arithmetic and high performance computing.
- Produced technical documentation presenting the theoretical foundation of the numerical results and sufficient explanation of methods for peer reproduction.
- Presented technical concepts in machine learning, deep learning, natural language processing and generative AI to both technical and non-technical audiences.
- Practiced proper data management of large, complex and HIPAA compliant datasets for local and HPC cluster analysis.
- Designed and delivered lectures, workshop tutorials, and presentations on introductory and intermediate college mathematics and data science in areas such as Natural Language Processing (NLP), Deep Learning, Large Language Models (LLMs).

Genentech

June 2023 – Aug. 2023

Early Clinical Development AI Intern

South San Francisco, CA

- Conducted applied AI research in digital pathology and bioinformatics using high-resolution medical imaging dataset from large data resources.
- Developed, trained and optimized deep learning and graph neural network models in Pytorch for survival prediction.
- Executed large model training on GPU enabled HPC clusters using slurm.
- Presented research findings to an interdisciplinary group of researchers.

Lightsense Inc

June 2022 – Dec. 2022

Research and Development Intern

Tucson, AZ

- Designed and trained predictive models and generative AI for data augmentation, outlier detection and classification of spectral data using scikit-learn and tensorflow.
- Performed exploratory data analysis, data visualization and model evaluation to improve performance of predictive models.
- Researched and presented modern papers in the intersection of machine learning and chemometrics.
- Collaborate with researcher as a mathematics and machine learning expert at a small and diverse start up.

🔧 SKILLS

- **Programming:** Python, Matlab, R, SQL, C
- **Machine Learning:** PyTorch, Tensorflow, Scikit-Learn, Deep Learning, NLP, Statistical Machine Learning.
- **Data Science:** Pandas, NumPy, Matplotlib, Optuna, MLFlow, Data Cleaning, Feature Engineering, Exploratory Data Analysis.
- **Software Engineering:** Git/Github, Linux, Docker, Object-oriented Programming, Unit testing, CI/CD, HPC, AWS (EC2, S3), Slurm.
- **Math and Statistics:** Experiment Design, Hypothesis Testing, Causal Inference, Optimization Techniques
- **Relevant Coursework:** Statistical Machine Learning, Algorithms of Applied Math, Statistical Experiment Design, Advanced Regression Analysis, Probabilistic Graphical Models

🔪 PORTFOLIO PROJECTS

Production MLOps Uplift Modeling and Causal Inference

June 2026

[Link to Github Repo](#)

- Developed end-to-end machine learning repo for training, tuning, deploying and monitoring of uplift models using Criteo's uplift dataset.
- MLOps implemented via MLFlow, Optuna, Dash and FastAPI.
- Production ready via FastAPI access point and data drift detection with triggered retraining.

EDUCATION

University of Arizona

PhD Mathematics

Aug. 2019 – Aug. 2025

Tucson, AZ

Western Washington University

MS Mathematics

Sept. 2016 – June. 2018

Bellingham, WA

University of Puget Sound

BS Mathematics

Aug. 2011 – May. 2015

Tacoma, WA